NanoGT Match Report: Spinal Muscular Atrophy

Disease: Spinal Muscular Atrophy (ORPHA:70)
Primary gene: SMN1
Gene CDS: 882 bp
Inheritance: Autosomal recessive
Target tissues: CNS, muscle

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	OAV101-IT	AAV9	9.5/10	High	approved
2	Zolgensma	AAV9	9.5/10	High	approved
3	AT132	AAV8	9.0/10	High	phase3
4	SRP-9001	AAV9	8.2/10	High	approved
5	DTX301	AAV8	6.8/10	Medium	phase2

Match #1: OAV101-IT

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/spinal cord
Composite score: 9.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	1.5	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 882bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: motor_neuron
- Approval status: approved

Match #2: Zolgensma

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/motor neuron
Composite score: 9.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	1.5	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 882bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: motor_neuron
- Approval status: approved

Match #3: AT132

Precedent disease: X-linked myotubular myopathy
Vector: AAV8
Tissue target: muscle
Composite score: 9.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	1.5	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.8	1.0

Rationale

- Gene CDS 882bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: muscle
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement

- Pathway match: myopathy
- Approval status: phase3

Match #4: SRP-9001

Precedent disease: Duchenne muscular dystrophy
Vector: AAV9
Tissue target: muscle
Composite score: 8.2 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	0.5	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 882bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: muscle
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Pathway match: myopathy
- Approval status: approved

Match #5: DTX301

Precedent disease: Ornithine transcarbamylase deficiency
Vector: AAV8
Tissue target: liver
Composite score: 6.8 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.5	2.0
Protein class	1.5	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.6	1.0

Rationale

- Gene CDS 882bp / cargo 4700bp (19% utilized)

- Vector tropism overlap: cns, muscle
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (motor_neuron vs urea_cycle)
- Approval status: phase2